Junhee **Cho**

Graduate Researcher in Quantum Computing & Semiconductor Devices

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Education

POSTECH(Pohang University of Science and Technology) *Pohang, Korea*

M.S. in Electrical Engineering *Mar. 2023 - Feb. 2026*

- GPA: 4.11 / 4.3
- Supervisor: Prof. Moonjoo Lee
- Thesis: Construction of a Cryogenic Ion-Trap Quantum Computing System and Investigation of Light-Induced Charging Effects from Electrode Patches

Hongik University *Seoul, Korea*

B.S. in Electronic and Electrical Engineering *Mar. 2016 - Feb. 2023*

- GPA: 3.96 / 4.5
- Supervisor: Prof. Ho-Young Cha

Publications

- **Junhee Cho**, Sangsoo Han, Myunghun Kim, Keumhyun Kim, Youngil Moon, Hyegoo Lee, Yongha Shin, Moonjoo Lee, "Surface Photovoltage Dynamics in Oxide Patches on Trap Electrodes and Effects on Trapped Ions", *On preparation*
- Seongchan Bae, Myunghun Kim, **Junhee Cho**, Moonjoo Lee, Jae-Yoon Sim, "A 600-V Peak-to-Peak 65-dBc RF Signal Source for Trapped-Ion Quantum Computing", *IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II*, (2024)

Patents

Deep learning inference device for ion qubit detection, ion qubit detection system, and ion qubit detection method

Moonjoo Lee, Junhee Cho, Yongha Shin, Keumhyun Kim, Hyegoo Lee *Nov. 2025*

- Korean Patent Application No. 10-2025-0165113

A single-ion nano engine

Moonjoo Lee, Myunghun Kim, Junhee Cho *Oct. 2023*

- Korean Patent Application No. 10-2023-0139870
- U.S. Patent Application No. 18/744,460 (2025)

Selected Conferences

A Segmented-blade Trap and Oscillatory Motion of Trapped Ions *Okinawa, Japan*

Myunghun Kim, **Junhee Cho**, Sangsoo Han, Sehyeon Gwon, Hyegoo Lee, Keumhyun Kim and Moonjoo Lee *Sep. 2024*

Advanced Quantum Technologies for Trapped Ions (AQTTI), Poster presentation

Constructing Ytterbium Ion Trap System for Quantum Computing using Cryostat *Gyeongju, Korea*

Junhee Cho, Myunghun Kim, Sehyeon Gwon, Keumhyun Kim, Hyegoo Lee, Sangsoo Han and Moonjoo Lee *Feb. 2024*

31st Korean conference on semiconductors (KCS), Poster presentation

A segmented-blade trap with biasing rods *Hannover, Germany*

Myunghun Kim, **Junhee Cho**, Keumhyun Kim, Hyegoo Lee, Jungsoo Hong and Moonjoo Lee *Sep. 2023*

7th European conference on trapped-ion(ECTI), Poster presentation

Enhancement-mode GaN MOS-HFET with Integrated Clamp Circuit *Jeongseon, Korea*

Junhee Cho, Seungheon Shin and Ho-Young Cha *Feb. 2023*

30th Korean conference on semiconductors (KCS), Poster presentation

Others (Presenter: Junhee Cho)

- Cryogenic Ion Trap System Development Process (KPS Fall Meeting 2023), Poster presentation
- Effects of light-induced charging to trapped ions at cryogenic temperatures (KPS Fall Meeting 2024), Oral presentation
- Characterization of charging effects due to light in cryogenic temperatures (KCS 2025), Oral presentation

Honors & Awards

Domestic Awards

Feb. 2023 **On-site Poster Award**, The 30th Korean Conference on Semiconductors — *Jeongseon, Korea*

Feb. 2024 **On-site Poster Award**, The 31st Korean Conference on Semiconductors — *Gyeongju, Korea*

Nov. 2024 **POSTECH Presidential Award**, POSTECH Startup Competition — *Pohang, Korea*

Research Experiences

M.S. Researcher — *Pohang, Korea*

POSTECH Quantum Computing and Quantum Networks Lab. — *Mar. 2023 - Feb. 2026*

Summary: Construction of a cryogenic ion-trap system for quantum computing

- Investigated the nonlinear Duffing oscillator dynamics of a single $^{174}Yb^+$ ion
- Designed and fabricated a multi-layer ion trap chip for scalable quantum-computing experiments
- Constructed an ultra-high vacuum (UHV), electrical delivery, $^{40}Ca^+$ ion fluorescence imaging system within a 4 K cryogenic station
- Designed and built a stable laser system and optical paths for trapped-ion manipulation
- Developed a PyQt-based user interface (UI) to automate the experimental setup
- Trapped two $^{40}Ca^+$ ion qubits and compensated micromotion by narrowing the Lorentzian linewidth in 397 nm spectroscopy
- Characterized light-induced charging effects of oxide patches based on Surface Photovoltage (SPV) theory on trap electrodes and their impact on the ion-trapping potential ***(manuscript in preparation)***
- Developed a real-time ion-detection program based on a Faster R-CNN model to automatically infer ion number and positions from EMCCD images

Undergraduate student — *Seoul, Korea*

Hongik Univ. Advanced Semiconductor Technology Lab. — *Aug. 2021 - Dec. 2022*

Summary: Enhancement-mode operation of depletion-mode GaN HEMT by integrating with clamp circuit

- Modeled and analyzed AlGaN/GaN HEMT using TCAD (Silvaco Atlas)
- Converted the TCAD device model into a SPICE model using the BSIM4 model
- Integrated the HEMT model with a clamp circuit in LTspice to achieve normally-off operation and optimized switching speed
- Achieved high power conversion efficiency in a GaN-based DC-DC boost converter

Skills

Data Analysis & Deep Learning	Python, C, Matlab, QuTiP, Qiskit, PyTorch
Device Design and Simulation	TCAD (Silvaco Atlas & Athena), SPICE (BSIM4, LTspice, Pspice), MATLAB, Microwind3, Ansys Maxwell 3D
Vacuum & Cryogenic Systems	Operation of rotary, turbo, ion and NEG pumps; Ti-sublimation pumps; and 4 K cryostations
Optics & Laser Control	Fiber delivery, double-pass AOM setups, PID-based laser power and frequency stabilization
Material Analysis and Imaging	SEM & EDS, 3D Profiler
Control & Embedded Systems	PyQt (Experimental GUI), FPGA, High-resolution DAC, Arduino, Raspberry Pi
3D modeling	Autodesk Inventor, Rhino 8